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ORIGINAL ARTICLE

Alternative Models to Hodgkin–Huxley Equations

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Abstract The Hodgkin and Huxley equations have served as the benchmark model in electrophysiology since 1950s. But it suffers from four major drawbacks. Firstly, it is only phenomenological not mechanistic. Secondly, it fails to exhibit the all-or-nothing firing mechanism for action potential generation. Thirdly, it lacks a theory for ion channel opening and closing activation across the cell membrane. Fourthly, it does not count for the phenomenon of voltage-gating which is vitally important for action potential generation. In this paper, a mathematical model for excitable membranes is constructed by introducing circuit characteristics for ion pump exchange, ion channel activation, and voltage-gating. It is demonstrated that the model is capable of re-establishing the Nernst resting potentials, explicitly exhibiting the all-or-nothing firing mechanism, and most important of all, filling the long-lasting theoretical gap by a unified theory on ion channel activation and voltage-gating. It is also demonstrated that the new model has one half fewer parameters but fits significantly better to experiment than the HH model does. The new model can be considered as an alternative template for neurons and excitable membranes when one looks for simpler models for mathematical studies and for forming large networks with fewer parameters.

Keywords Ion pump · Channel activation · Voltage-gating · Neural circuit model · Action potential

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1 Introduction

When Hodgkin and Huxley introduced their model for the squid giant axon (Hodgkin and Huxley 1952a), they were fully aware of their model's drawbacks because it was only a phenomenological fit to their experimental data. They foretold specifically that different empirical forms should fit the same data better, and other researchers agreed (Agin 1963; Cole 1968; Agin 1972; Almers 1978). Alternative models were made (Agin 1963) but never gained any traction because there was no point to replace one *ad hoc* model by another arbitrary one. Because it was a phenomenological fit, the ion channel opening and closing activation of the cell membrane was left unexplained in theory. As pointed out in FitzHugh (1961) the action potential's firing mechanism of the HH model is not the all-or-nothing type but a graded type, inconsistent with empirical findings. The last major shortcoming of the model was not anticipated by its authors because the process it should model was only discovered a decade later. It is the phenomenon of voltage-gating (Armstrong and Bezanilla 1973) by which charged molecules from the sodium channel pores are released to the outside of the cell in responding to some conformational changes in the pores to depolarizing voltage and for the effect of keeping the membrane from depolarizing. These problems have not been addressed sufficiently in terms of updated models. Without doubt it is important and useful to have more accurate, simpler, and more mechanistic models as basic building blocks for neural networks because such models should help the field to build it up to the next levels of sophistication. By mechanistic it is meant for a model to be applicable for physical processes or objects of the same type. Newton's gravitation law is the first and one perfect example of mechanistic modeling because it applies to all macroscopic bodies of mass. Goldman's (1943) derivation of Nernst potentials across cell membranes is mechanistic because it applies to all ion species. In contrast Hodgkin and Huxley's treatments of the sodium and the potassium currents probably cannot be considered mechanistic because their hypothesis for one ion type does not apply for the other type.

The purpose of this paper is try to come up with a mechanistic alternative to neuron modeling. The idea is to model the membrane as a circuit of devices each is defined by a current–voltage characteristics. More specifically, we will model the sodium–potassium ion pump exchanger by the *IV* characteristics that the time-rate of change in the current is proportional to the power of the pump. We will model the ion channel activation and the voltage-gating by one unified *IV* characteristics that the voltage-rate of change in the conductance is proportional to the conductance. We will demonstrate that the resulting conductance adaptation model is capable of reproducing all known phenomena of the HH model and much more. We will also demonstrate that the new model is much simpler and more accurate than the HH model, and hence, provides such alternative models.

2 The Result

The approach to modeling the squid giant axon as an electrical circuit began with Cole's work (1932, 1940, 1949) that the axon membrane had a rather narrow range

around 1 μF/cm^2 for membrane capacitance. Let V be the intracellular potential difference (inside potential minus outside potential) and $I = C\mathrm{d}V/\mathrm{d}t$ denote the capacitance current. Let I_{K}, I_{Na} denote, respectively, the potassium and the sodium ion current using the outward direction as the reference direction. Rather than the leakage current considered by the HH model, we consider instead the voltage-gating current I_{G} measured outward. Last let I_{ext} be the net remaining current with the inward reference direction. Then, the following equation must be the first equation of any circuit model of the membrane by Kirchhoff's Current Law

$$C\frac{\mathrm{d}V}{\mathrm{d}t} = -[I_{\mathrm{K}} + I_{\mathrm{Na}} + I_{\mathrm{G}} - I_{\mathrm{ext}}].$$

The aim of this section is to derive the functional forms for the potassium, the sodium, and the gating currents based on our proposed mathematical models for ion pump exchange, ion channel activation, and voltage-gating, and to obtain the resulting model for the membrane.

2.1 Ion Pump

Let q denote the intracellular charge difference (charge concentration inside minus outside) of a given ion species across the cell membrane. The following as a mathematical model for the pump was proposed in Deng (2009)

$$\frac{\mathrm{d}^2q}{\mathrm{d}t^2} = \phi\frac{\mathrm{d}q}{\mathrm{d}t}q \tag{1}$$

with ϕ a parameter. If we let $I = \mathrm{d}q/\mathrm{d}t$ be the current through the ion pump and $V = q/C$ be the across-membrane potential over the pump with C being the membrane capacitance. Then, the model can be interpreted as the IV characteristics of the pump as an electrical device because the model is equivalent to

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \lambda IV$$

with $\lambda = C\phi$ referred to as the pump parameter in the unit of per time per voltage. As the product IV represents power, the pump characteristics can be stated as the rate of change in the ion pump current is proportional to the power of the pump. This is a way to model the energy transfer of the pump when ATP is converted to ADP in exchange for ion transportation across the membrane.

The minimal circuit consisting a pump, a resistor, and a capacitor in loop can be understood completely, c.f. Fig. 1. The ordinary differential equations for the circuit are

$$CV' = I, \quad I' = -\lambda I(V + \gamma I). \tag{2}$$

Eliminating the time variable, the equation becomes $\mathrm{d}I/\mathrm{d}V = -\lambda C(V + \gamma I)$ which being a linear equation can be solved explicitly. In fact all important insights can be

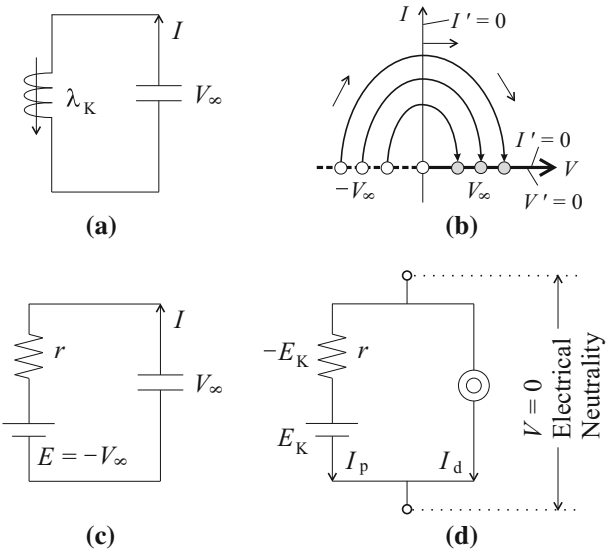


Fig. 1 **a** Circuit diagram of the potassium ion pump with the capacitive membrane. **b** The phase plane portrait of the circuit equations (2) with $\gamma = 0$. **c** A *CRE*-circuit approximation of the pump circuit. **d** The Nernst potential under electrical neutrality and across-membrane diffusion

obtained by dropping the resister term ($\gamma = 0$), for which the solution with the initial condition $V(0) = 0$, $I(0) = I_0 \geq 0$ is

$$V(t) = V_\infty \frac{1 - \mathrm{e}^{-\lambda V_\infty t}}{1 + \mathrm{e}^{-\lambda V_\infty t}} \quad \text{and} \quad I(t) = 2\lambda C V_\infty^2 \frac{\mathrm{e}^{-\lambda V_\infty t}}{[1 + \mathrm{e}^{-\lambda V_\infty t}]^2} \tag{3}$$

with

$$V_\infty = \sqrt{\frac{2I_0}{\lambda C}}.$$

The solution for the current means the pump is always unidirectional. That is, if it is positive at one point, say $I_0 > 0$, then it stays positive all the time. Equally important, as $t \to -\infty$, $\lim V(t) = -V_\infty$, $\lim I(t) = 0$ and as $t \to +\infty$, $\lim V(t) = V_\infty$, $\lim I(t) = 0$. This means if the membrane is negatively charged in the past ($\sim -V_\infty < 0$) then the membrane will be positively charged eventually ($\sim V_\infty > 0$) if the pump's transporting direction is from outside to inside as with the case of the potassium pump. Similarly, for the sodium pump, it will transport all sodium ion from inside to outside in asymptote.

We demonstrate next that the minimal pump–capacitor circuit is exponentially close to a resistor–battery–capacitor circuit in loop with the battery value given by $E = -V_\infty$, c.f. Fig. 1. The differential equation for the circuit is: $CV' = I = -(V + E)/r$ with the resistance r to be determined. Solving it with the zero voltage initial condition

$V(0) = 0$ as we did for the pump circuit, we have

$$V(t) = -E[1 - \mathrm{e}^{-t/(rC)}] = V_\infty[1 - \mathrm{e}^{-t/(rC)}].$$

So if we choose the resistance as

$$r = \frac{1}{\lambda C V_\infty} \tag{4}$$

then the dynamics for the battery circuit and the pump circuit are exponentially close for large t since the pump dynamics is approximately $V(t) = V_\infty[1 - \mathrm{e}^{-\lambda V_\infty t}]/[1 + \mathrm{e}^{-\lambda V_\infty t}] \sim V_\infty[1 - \mathrm{e}^{-t/(rC)}]$, converging to the same resting potential V_∞ at the same exponential rate $\mathrm{e}^{-\lambda V_\infty t}$, and starting from the same initial value $V(0) = 0$. That is, the ion pump can be approximated by a conductor-battery in series with the conductance $1/r = \lambda C V_\infty$. This approximation becomes more accurate as time goes by. This gives a mechanistic justification for Hodgkin and Huxley's modeling of ion channel currents by battery-driven ion channels.

2.2 Nernst Potential

The pump dynamics above says that in the absence of other forces such as diffusion a particular ion pump will eventually deposit all ions of the same type from one side of the membrane to the other side. We determine next the value of the asymptotic potential V_∞ of the pump under the influence of diffusion and the electrical neutrality condition of the membrane, c.f. Fig. 1, which is an illustration for the potassium ion. By Bernstein's (1912) hypothesis, this inward current due to the electromotive force is balanced out by the outward diffusion, assuming the membrane is permeable to potassium ions. Specifically, let I_p, I_d denote the pump current and the diffusive current as shown. Let the reference direction be inward and be defined by the across-membrane variable x from outside to inside and a be the thickness of the membrane. Let $n(x)$ be the ion density at position x and let $E_\mathrm{K} = -V_\infty$ be the equilibrium state reached when these two currents cancel each other out under the neutrality assumption $V = 0$. Then, the same Bernstein–Goldman equation below must hold (Goldman 1943; Hodgkin and Katz 1949)

$$0 = I_d + I_p = D\left(-\frac{dn}{dx}\right) + \mu n\left(-\frac{E_\mathrm{K}}{a}\right),$$

with D the diffusion constant and μ the mobility parameter (Hodgkin and Katz 1949). Solving this linear equation in n from $x = 0$ to $x = a$, and then expressing the result in E_K to obtain the following which is exactly the potassium ion's Nernst resting potential

$$E_\mathrm{K} = -\frac{D}{\mu} \ln \frac{[\mathrm{K}]_i}{[\mathrm{K}]_o}$$

with $[\mathrm{K}]_i = n(a)$, $[\mathrm{K}]_o = n(0)$ denoting the inside and outside potassium concentration, respectively. The same derivation can be used to obtain the sodium ion's Nernst potential.

2.3 Ion Channel Activation

With the presence of various types of ions and charged molecules, the across-membrane potential V is the manifestation of their aggregate. The independence hypothesis in electrophysiology holds that the opening and closing of an ion gate is a function of the voltage not of the other ions. As a consequence individual ion currents are modeled by the ohmic IV characteristics form $I = g(V - E)$. By eliminating the spatial effect of axon by their voltage-clamp experiments Hodgkin and Huxley showed (Hodgkin et al. 1952; Hodgkin and Huxley 1952b, c, d) the conductance g undergo changes throughout the course of an action potential and had the insight to propose it to be a function of the voltage, that is, $g = g(V)$. Their work showed that throughout the course of an action potential the ion conductances for sodium and potassium ions, g_{Na}, g_{K}, increase from near zero when the membrane is at rest and then decrease to near zero again when the action potential ends, hence implying that there are ion channels which open to increase their conductances and close to decrease them. Here is where the HH model becomes empirical rather than mechanistic because of its use of arbitrary functional forms for the sodium and potassium channel conductances.

By channel activation, we mean channel opening and closing. We propose the following model for both types of activation

$$\Delta g \sim g \Delta V \quad \text{equivalently} \quad \frac{\mathrm{d}g}{\mathrm{d}V} = \frac{g}{b} \tag{5}$$

with $b > 0$ the activation parameter in the dimensional unit of voltage. This means V is increasing ($\Delta V > 0$) if and only if the conductance g is increasing ($\Delta g > 0$), and the rate of change in the conductance with respect to the intracellular voltage is proportional to the conductance. It is a dual feedback process—positive feedback with increasing voltage and negative feedback with the opposite. Solving it we get the exponential activation law $g(V) = a\mathrm{e}^{V/b}$. Normalizing it at the ion's Nernst potential with $g(E) = \bar{g}$ we get

$$g(V) = \bar{g}\mathrm{e}^{(V-E)/b}.$$

Namely $a = \bar{g}\mathrm{e}^{-E/b}$. The parameter $g(E) = \bar{g}$ is referred to as the intrinsic conductance. It is a simple exercise to check that the corresponding IV-curve $I = f(V) = \bar{g}\mathrm{e}^{(V-E)/b}(V - E)$ has a unique minimal point at $\bar{b} = E - b$ whose value is $-\bar{g}b\mathrm{e}^{-1}$, c.f. Fig. 2. To the right side of $\bar{b}$, the IV-curve f is increasing, crossing the V axis at the only resting potential E. To the left side of the minimum criticality $\bar{b}$, f is decreasing, giving rise to a varying negative conductance. It approaches $I = 0$ asymptotically from below and at an exponential rate as V decreases. For reasons which will become clearer later, we will refer to parameter b as the activation range parameter, measuring the minimal current point from the Nernst potential.

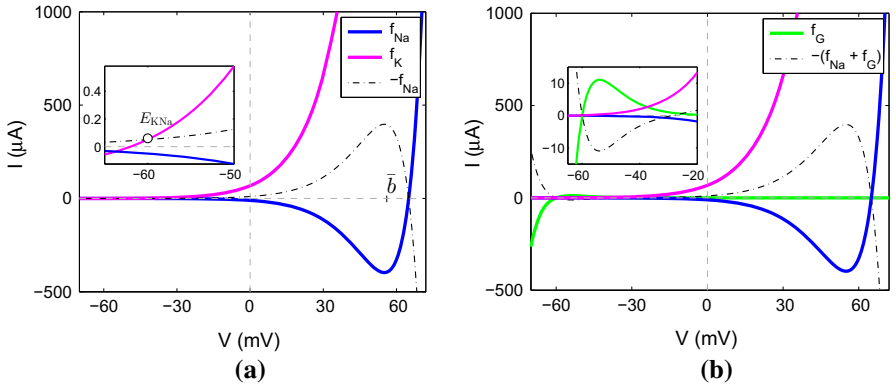


Fig. 2 (Color figure online) **a** Typical *IV* characteristics curves for the sodium, potassium, f_{Na}, f_{K}, respectively. The left-most intersection of $I = f_{\mathrm{K}}$ and $I = -f_{\mathrm{Na}}(V)$, E_{KNa}, is a first approximation of the membrane resting potential E_m. But the geometric configuration of the two *IV*-curves implies the KNa-equilibrium state E_{KNa} is almost always unstable. Here $\bar{b} = E_{\mathrm{Na}} - b_{\mathrm{Na}}$ is the minimum point of the f_{Na}-curve. **b** A similar plot as **a** except for the addition of the gating *IV*-curve f_{G} with $E_{\mathrm{G}} = E_{\mathrm{KNa}}$. The true resting membrane potential E_m is the left-most intersection of $I = f_{\mathrm{K}}(V)$ and $I = -(f_{\mathrm{Na}}(V) + f_{\mathrm{G}}(V))$, as shown in the *inset*

The proposed activation characteristics have this property of dichotomy. If the intracellular membrane voltage is increasing in time, $\mathrm{d}V/\mathrm{d}t > 0$, then the activation gates open up exponentially fast in conductance as $\mathrm{d}g/\mathrm{d}t = \mathrm{d}g/\mathrm{d}V \cdot \mathrm{d}V/\mathrm{d}t = kg$ with $k = (\mathrm{d}V/\mathrm{d}t)/b > 0$. In contrast, if the membrane voltage is decreasing in time, $\mathrm{d}V/\mathrm{d}t < 0$, then the activation gates close down exponentially fast as well for $\mathrm{d}g/\mathrm{d}t = kg$ with $k = (\mathrm{d}V/\mathrm{d}t)/b < 0$. It is a model for both opening activation and closing activation with the same voltage-specific exponential rate. That is, opening gates beget more gates opened and closing gates triggers more gates closed.

For the potassium and sodium *IV* characteristics we get, respectively,

$$f_{\mathrm{K}}(V) = \bar{g}_{\mathrm{K}} \mathrm{e}^{(V-E_{\mathrm{K}})/b_{\mathrm{K}}}(V - E_{\mathrm{K}}), \quad f_{\mathrm{Na}}(V) = \bar{g}_{\mathrm{Na}} \mathrm{e}^{(V-E_{\mathrm{Na}})/b_{\mathrm{Na}}}(V - E_{\mathrm{Na}}).$$

In the operating range of action potentials between the potassium and sodium Nernst potentials, $(E_{\mathrm{K}}, E_{\mathrm{Na}})$, only the critical potential $E_{\mathrm{Na}} - b_{\mathrm{Na}}$ of the sodium current may lie, predicting that the potassium current is always outward and the sodium current is always inward which in turn has two different phases: increased entry into the cell to the left of the critical point $E_{\mathrm{Na}} - b_{\mathrm{Na}}$ and decreased entry to the right of the criticality. That is, inside this operating range the sodium and potassium *IV*-curves behave differently qualitatively, the former usually have a negative conductance branch but the latter's conductance is always positive.

There is a quantitative difference between the intrinsic conductances $\bar{g}_{\mathrm{K}}$ and $\bar{g}_{\mathrm{Na}}$. First we know from the derivation of the Nernst potential above [specifically (4)] that the conductance is

$$g_{\mathrm{K}}(0) = \bar{g}_{\mathrm{K}} \mathrm{e}^{-E_{\mathrm{K}}/b_{\mathrm{K}}} = \frac{1}{r} = \lambda C V_{\infty}$$

with $V_\infty \sim |E_\mathrm{K}|$. There is no need for a precise value of V_∞, an estimate of its range or average of its range suffices. Say $E_\mathrm{K} \sim -60$ and $V_\infty \sim 60$. This gives rise to an estimate of

$$\bar{g}_\mathrm{K} \sim \lambda C 60 \mathrm{e}^{E_\mathrm{K}/b_\mathrm{K}} = \lambda C 60 \mathrm{e}^{-60/b_\mathrm{K}}.$$

Similarly, if we use $E_\mathrm{Na} \sim +70$, a similar estimate is obtained for the sodium conductance:

$$\bar{g}_\mathrm{Na} \sim \lambda C 70 \mathrm{e}^{E_\mathrm{Na}/b_\mathrm{Na}} = \lambda C 70 \mathrm{e}^{70/b_\mathrm{Na}}.$$

The conclusion is if the b-parameter values for both ions are comparable, say in the decade range, then relative to each other, the intrinsic potassium conductance $\bar{g}_\mathrm{K}$ should be a few orders smaller than the intrinsic sodium conductance $\bar{g}_\mathrm{Na}$. From the HH model, the intrinsic potassium conductance is worked out to be $\bar{g}_\mathrm{K} = 0.0229$ m mho/cm^2. As a result a guessed value is used for the intrinsic sodium conductance $\bar{g}_\mathrm{Na} = 100$ m mho/cm^2, both are used below and for the illustrations of the *IV*-curves in Fig. 2.

2.4 Voltage-Gating

A major advance in neurophysiology was made in the 1970s (Armstrong and Bezanilla 1973) by the discovery of the voltage-gating phenomenon; whereby, there is a small pulse-like outward current opposite to the inward entry of sodium ions during the onset of action potential across the squid giant axon membrane when the membrane is depolarizing. The gating current was not recognized by Hodgkin and Huxley but was nonetheless captured by their meticulously fitted model. For the *IV*-curve of the gating current, we propose first it has a resting potential E_G, which may be taken to be the intersection of the sodium and potassium *IV*-curves, calling it the provisional resting potential E_KNa, and second, it is nonlinear ohmic of a similar form as for channel activation $I = f_\mathrm{G}(V) = g_\mathrm{G}(V)(V - E_\mathrm{G})$ except for a negative proportionality:

$$\frac{\mathrm{d}g}{\mathrm{d}V} = -\frac{g}{b} \tag{6}$$

with $b > 0$. That is, positive proportionality is for activation and negative proportionality is for gating. Gating conductance recedes in proportion to itself with depolarizing voltage ($\Delta g < 0$ if and only if $\Delta V > 0$). Solving it the corresponding *IV*-curve becomes

$$I = f_\mathrm{G}(V) = \bar{g}_\mathrm{G} \mathrm{e}^{-(V-E_\mathrm{G})/b_\mathrm{G}} (V - E_\mathrm{G}).$$

For this type of function, it has the unique maximum point at the critical voltage $E_\mathrm{G} + b_\mathrm{G}$ right of the gating equilibrium E_G. For $E_\mathrm{G} < V < E_\mathrm{G} + b_\mathrm{G}$, the gating current is increasing in V and outward, preventing the membrane from depolarization. Its maximum defines the minimum threshold for excitation current to clear in order

for action potentials to generate. For this reason, parameter b_{G} is referred to as the gating range parameter, similar to the activation range parameters b_{K}, b_{Na}.

Notice that once the threshold is cleared and the membrane is depolarizing (below $V = 0$ and $\mathrm{d}V/\mathrm{d}t > 0$), the gating conductance drops exponentially fast with $\mathrm{d}g/\mathrm{d}t = \mathrm{d}g/\mathrm{d}V \cdot \mathrm{d}V/\mathrm{d}t = -kg$ with $k = (\mathrm{d}V/\mathrm{d}t)/b > 0$, permitting the generation of an action potential. When the membrane is hyperpolarizing (below $V = 0$ and $\mathrm{d}V/\mathrm{d}t < 0$), gating becomes active again with the conductance approaching the intrinsic conductance $\bar{g}_{\mathrm{G}}$. To the left side of the gating reversal potential, $V < E_{\mathrm{G}}$, the gating current $I = f_{\mathrm{G}}(V)$ is negative (inward), effectively shutting down hyperpolarization and restoring the membrane to its resting state at the same time.

In addition, the sodium-gating parallel combination *IV*-curve, $I = f_{\mathrm{Na}}(V) + f_{\mathrm{G}}(V)$, is typically N-shaped. The left knee point of the N characteristics is near the gating criticality $E_{\mathrm{G}} + b_{\mathrm{G}}$, and the right knee point is near the sodium criticality $E_{\mathrm{Na}} - b_{\mathrm{Na}}$. This is because the tail part of each curve beyond its critical point is exponentially flat. Furthermore, the membrane resting potential E_{m} is the intersection of the potassium *IV*-curve $I = f_{\mathrm{K}}(V)$ and the reflection of the sodium-gating *IV*-curve $I = -(f_{\mathrm{Na}}(V) + f_{\mathrm{G}}(V))$ (so that $f_{\mathrm{K}}(E_{\mathrm{m}}) + f_{\mathrm{Na}}(E_{\mathrm{m}}) + f_{\mathrm{G}}(E_{\mathrm{m}}) = 0$). It is to the right side of E_{G} and below E_{Na}, and is always stable, see Fig. 2. Negative conductance has always been a theoretical conundrum (Hodgkin and Huxley 1952c; Moore 1959; FitzHugh 1961; Yamamoto 1965; Agin 1972). For our model, it is a mechanistic consequence to channel activation (5) and voltage-gating (6).

2.5 Conductance Adaptation and Circuit Model

Although each current's *IV*-curve is accessible as an equilibrium current for each clamped voltage, in transient the voltage driving conductance $g(V) = \bar{g}\mathrm{e}^{\pm(V-E)/b}$ cannot be realized instantaneously during action potential generation and propagation. There is a time delay or time-course adaptation. We propose the following conductance adaptation following the idea of Hodgkin and Huxley (1952a):

$$\frac{\mathrm{d}p}{\mathrm{d}t} = \tau(\mathrm{e}^{\pm(V(t)-E)/b} - p) \tag{7}$$

with τ being a time constant. Thus, instead of the *IV*-curve the corresponding current's time course is given by $I(t) = \bar{g}\,p(t)(V(t)-E)$ with $p(t)$ determined by the adaptation equation above (Fig. 3).

Putting all these assumptions together, we obtain the following mathematical model for the squid giant axon and excitable membranes in general:

$$\begin{cases} CV' = -[\bar{g}_{\mathrm{K}} n(V - E_{\mathrm{K}}) + \bar{g}_{\mathrm{Na}} m(V - E_{\mathrm{Na}}) + \bar{g}_{\mathrm{G}} h(V - E_{\mathrm{G}}) - I_{\mathrm{ext}}] \\ n' = \tau_{\mathrm{K}}(\mathrm{e}^{(V-E_{\mathrm{K}})/b_{\mathrm{K}}} - n) \\ m' = \tau_{\mathrm{Na}}(\mathrm{e}^{(V-E_{\mathrm{Na}})/b_{\mathrm{Na}}} - m) \\ h' = \tau_{\mathrm{G}}(\mathrm{e}^{-(V-E_{\mathrm{G}})/b_{\mathrm{G}}} - h) \end{cases} \tag{8}$$

where I_{ext} denotes any intracellular current other than the potassium, the sodium, the gating, and the capacitive currents. As for the conductance adaptation time constants,

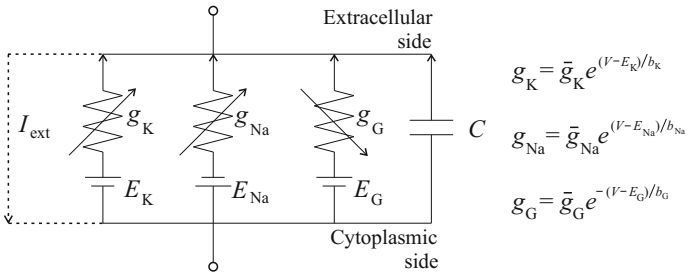


Fig. 3 A circuit representation of model (8) of which a voltage-gating current replaces the leakage current in the HH model. The activation conductance and the gating conductance are of the same mathematical form but exactly opposite in the direction of the membrane potential

we may assume the sodium channel and the gating channel share the same constant $\tau_{Na} = \tau_G = \tau_{NaG}$ because they are structurally bonded together. The dimensionless variables n, m, h are not the same as of the HH model but are borrowed here to pay tribute to Hodgkin and Huxley's seminal work.

2.6 Best Fit of Model to Data for Parameter Estimation

A piece of mathematics only remains as a conceptual model for any physical process unless and until it is best fitted to the process to fix a parameter point or a parameter range for the model. Otherwise different qualitative behaviors of the conceptual model would forever remain as unrealized and untested possibilities. To this end, we re-sampled the action potential data of axon 17 from Fig.12 of Hodgkin and Huxley (1952a) and fitted our model to the data to see how well or badly the model performs.

First we fix some parameter values and exclude them from the best fit process because they are known. These are: the capacitance C, the two ions Nernst resting potentials E_K, E_{Na}, the intrinsic potassium conductance $\bar{g}_K$. The membrane capacitance is $C = 1\,\mu F/cm^2$. The Nernst potentials are extracted from Hodgkin and Huxley (1952a) which had them shifted up by the Goldman–Hodgkin–Katz potential E_r which was estimated by Hodgkin and Katz (1949) to be around -47.5 mV, leading to $E_K = -59.5$ mV and $E_{Na} = 67.5$ mV, respectively. (As a result, the HHAxon17 data are shifted down by the supposedly membrane resting potential E_r.) As for the intrinsic potassium conductance $\bar{g}_K$, it is the value of $g_K(E_K)$. As mentioned above, it can be worked out to be $\bar{g}_K = 0.0229$ m mho/cm^2 from Hodgkin and Huxley (1952a).

The rest parameters are each given an estimated starting value and then best fitted by a gradient search method. First we cannot do the same for the intrinsic sodium conductance $\bar{g}_{Na}$ as we did for the intrinsic potassium conductance $\bar{g}_K$ because we know now that Hodgkin and Huxley's purportedly sodium *IV*-curve is an aggregate of the sodium and the gating channels. The estimate for the potassium conductance is reliable but not for the sodium conductance. Because of our estimation above on the order of magnitude for the intrinsic conductances of the potassium and sodium channels, we will choose a starting value for the latter to be $\bar{g}_{Na} = 100$ m mho/cm^2.

As for the gating parameter, we start the initial guess for the gating resting potential at $E_{\rm G} = -52.5$, 5 mV below the estimated membrane resting potential $E_r = -47.5$ mV. As for the intrinsic gating conductance, we will start it at $\bar{g}_{\rm G} = 10$ m mho/cm^2, a value between $\bar{g}_{\rm K}$ and $\bar{g}_{\rm Na}$.

As for the activation and gating range parameters, we will start them at $b_{\rm Na} = 10$ mV, $b_{\rm K} = 10$ mV, $b_{\rm G} = 8$ mV. This translates to the reversal of activation for the sodium channel at $E_{\rm Na} - b_{\rm Na} = 49.5$ mV, a non-accessible and thus unimportant potassium criticality $E_{\rm K} - b_{\rm K} = -69.5$ mV, and a gating 'threshold' criticality $E_{\rm G} + b_{\rm G} = -44.5$ mV.

Last, for the adaptation time constants we use $\tau_{\rm K} = 1$/msec, and $\tau_{\rm Na} = 10$/msec, making the sodium conductance adaptation one order faster than the potassium conductance. This is based on Hodgkin and Huxley's estimate that potassium channels

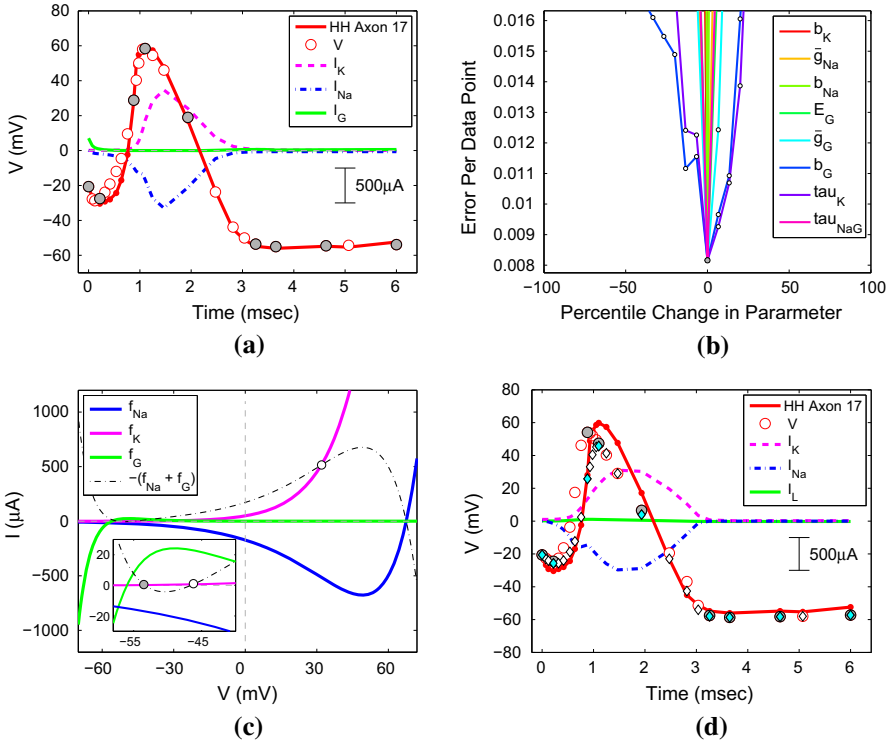


Fig. 4 (Color figure online) **a** Following the initial guess from the text, the best fitted parameter values for the circuit model (8) are: $E_{\rm K} = -59.5$ mV, $\bar{g}_{\rm K} = 0.0229$ m mho/cm^2, $b_{\rm K} = 16.6$ mV, $E_{\rm Na} = 67.5$ mV, $\bar{g}_{\rm Na} = 100$ m mho/cm^2, $b_{\rm Na} = 18.4$ mV, $E_{\rm G} = -56$ mV, $\bar{g}_{\rm G} = 9.3333$ m mho/cm^2, $b_{\rm G} = 7.0667$ mV, $C = 1\mu$F/cm^2, $\tau_{\rm K} = 0.8667$/msec, and $\tau_{\rm NaG} = 10$/msec. Filled data points are used for the second-order error function. **b** The per-data-point error function in each searched parameter centered at the best fitted point whose error value is 0.0082. **c** *IV*-curves of the best fitted model, showing three equilibrium states with the left-most being the resting membrane potential E_m. **d** A similar plot for the original fit of the HH model to the same data, and a best fit showing the voltage trace only (*diamond marker*) starting at the original parameter values of the HH model with searching parameters not shared with our model (8). The original fit error is 0.0595, and the best fit error is 0.0395, both are worse than the best fit of our model

open and close roughly 10 times slower than sodium channels (Hodgkin and Huxley 1952a; Almers 1978).

The HHAxon17 data were obtained by Hodgkin and Huxley by an instantaneous depolarization above the resting potential of the axon that translates to the initial voltage value $V(0) = -20.6707$ mV. The initial values for the rest of variables are $n(0) = \mathrm{e}^{(E_r - E_\mathrm{K})/b_\mathrm{K}}$, $m(0) = \mathrm{e}^{(E_r - E_\mathrm{Na})/b_\mathrm{Na}}$, $h(0) = \mathrm{e}^{-(E_r - E_\mathrm{G})/b_\mathrm{G}}$, which are fixed throughout the best fit process because they are good approximations for all searches at the initial time $t = 0$.

A best fit of the model to the data is done by Newton's line search method (c.f. Deng et al. 2014; Deng 2014). The error function is defined to be

$$\mathcal{E} = \frac{1}{N}\sqrt{\sum_{j=1}^{N}\left[\frac{|V(t_i) - V_i|}{\bar{V}}\right]^2}$$

where $\bar{V}$ is the maximum of the absolute values of the voltage data, N is the number of data points included for the error function. Thus, $\mathcal{E}$ measures the per-data-point relative error between the predicted value $V(t_i)$ by the model and the observed value V_i at time t_i. Fitted points include only these: the end points, the maximal and the minimal points, and the inflection points, constituting the so-called second-order fit. (The first-order fit by definition would exclude the inflection points of the data.) The line search is to find the so-called best fitted parameter values from the starting parameter values that gives rise to a local minimum point of the error function $\mathcal{E}$. At any iteration of the search, the error is calculated at a discrete set of points from an interval of each parameter and the new starting parameter point is chosen if it defines the smallest error. The interval is centered at the parameter value with the radius of the absolute value of the parameter, i.e., either $[0, 2p]$ or $[2p, 0]$ depending on if the current parameter value p is positive or negative. Figure 4 shows both a best fit of the model and the original and a best fit of the HH model to the same data. This preliminary comparison suggests our model does better than the HH model.

3 Discussion

Mathematical modeling is a process of refinement. Its aim is always to find better approximations of natural processes. In addition to the differences and similarities discussed in the modeling passages above, more comparisons between the two models are given as follows.

3.1 Absolute Refraction and Anode Break Excitation

The HH model was hugely successful. Its success was supported among other things by its matching up two uncanny properties of Hodgkin and Huxley's experiment data, the phenomenon of absolute refraction and the anode break excitation oscillation. The former occurs when a sudden initial depolarizing voltage is applied the membrane voltage decreases first before increasing or depolarizing as shown by both the axon

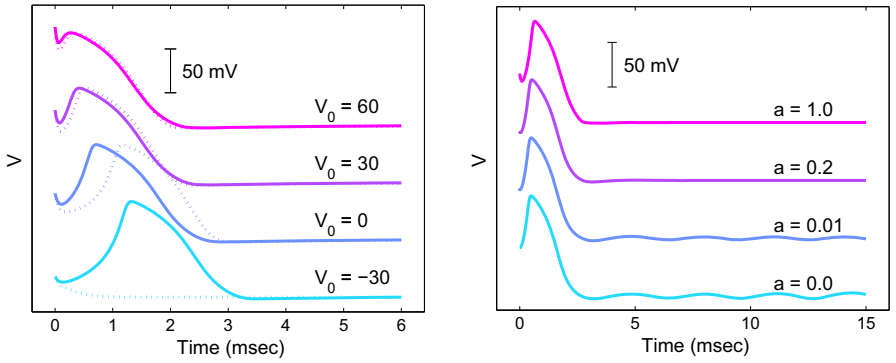


Fig. 5 (Color figure online) **a** Phenomenon of the so-called absolute refraction by Hodgkin and Huxley (1952a) for different depolarizing initial voltage with the same parameter values from Fig. 4 for model (8). *Dotted curves* are the same plot except for a larger intrinsic conductance $\bar{g}_G = 15$ for gating. **b** Anode break excitation oscillations for small $0 < a < 1$ for equations (9) with the same parameter values as **a** with $\bar{g}_G = 15$ except for the new additions a and $\tau_G = 0.5$

17 data and the models in Fig. 4. This phenomenon was later identified to be the phenomenon of voltage-gating (Armstrong and Bezanilla 1973). It is validated by our model (8) as demonstrated in Fig. 5a. It shows not only model (8) exhibits the same property but also its affect by the intrinsic gating conductance $\bar{g}_G$ as it is supposed to be. In contrast, the HH model exhibits the same but does not explain it.

As for the anode break excitation, it refers to the phenomenon that after hyperpolarization and during the phase of repolarization, the membrane voltage oscillates toward its resting potential rather than does so monotonically. The HH model is capable of this phenomenon but does not match up well in the timescale (Fig.22 of Hodgkin and Huxley 1952a). To capture this property with the correct timescale, we make a small modification to model (8). We assume that a proportion of the gating current $I_G = \bar{g}_G e^{-(V-E_G)/b_G}(V - E_G)$ is subject to conductance adaptation and the remaining proportion is subject to current adaptation modeled by the last equation of the system below

$$\begin{cases} CV' = -[\bar{g}_K n(V - E_K) + \bar{g}_{Na} m(V - E_{Na}) \\ \qquad + a\bar{g}_G h(V - E_G) + (1 - a)I_G - I_{ext}] \\ n' = \tau_K (e^{(V-E_K)/b_K} - n) \\ m' = \tau_{NaG} (e^{(V-E_{Na})/b_{Na}} - m) \\ h' = \tau_{NaG} (e^{-(V-E_G)/b_G} - h) \\ I_G{}' = \tau_G (\bar{g}_G e^{-(V-E_G)/b_G}(V - E_G) - I_G) \end{cases} \tag{9}$$

where $a : (1 - a)$ is the proportionality split for the two types of adaptation. Figure 5a shows that if the proportion for current adaptation is substantial (small a) but slow (small τ_G), anode break excitation occurs with the same timescale as shown by the experimental data of Hodgkin and Huxley (1952a) (Fig.22). We should note that the phenomena of absolute refraction and anode break excitation are notorious for

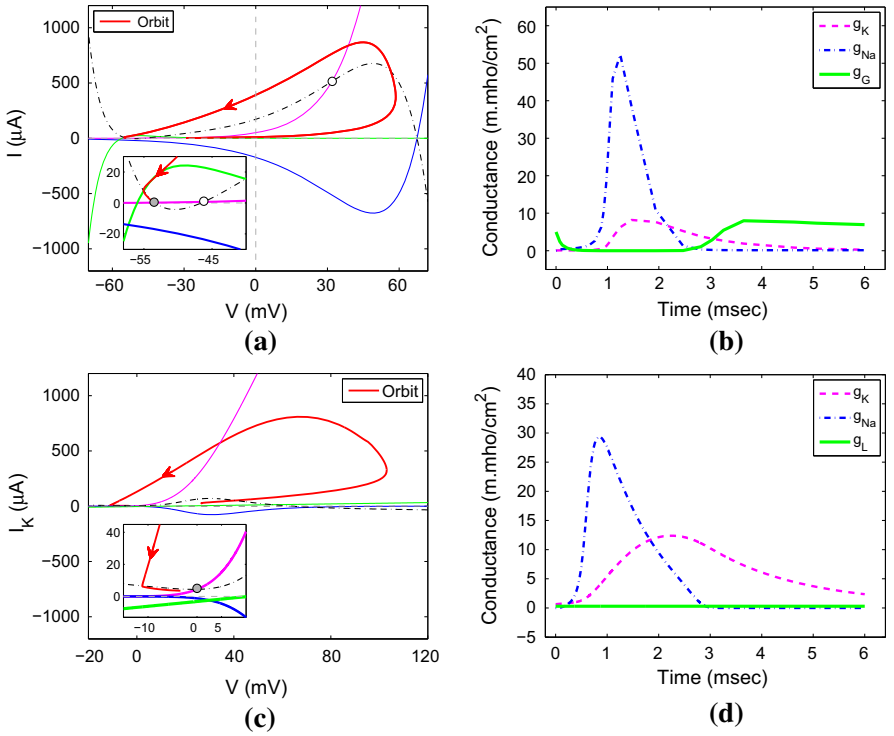


Fig. 6 (Color figure online) **a** Best fitted action potential of (8) (labeled as orbit) is projected to the VI_{K}-space together with its various IV-curves the same as Fig. 4c. **b** Its conductances as functions of the time for the action potential. **c** and **d** Similar plot as **a** and **b** for the HH model. The *dash-dot curve* is the reflection of the sodium-leakage IV-curve $I = -(f_{\mathrm{Na}}(V) + f_{\mathrm{L}}(V))$ in **c**

other models to replicate, the FitzHugh–Nagumo model (FitzHugh 1961) and the Hindmarsh–Rose model (Hindmarsh and Rose 1984) are two such examples.

3.2 All-or-Nothing Action Potentials

Although both the HH model and our model (8) match up Hodgkin and Huxley's experimental data in many aspects, many fundamental differences remain. Figure 6 shows some. The first concerns the all-or-nothing generation of action potentials. As it is shown in Fig. 6a, in addition to the membrane resting potential E_{m}, our model has two more equilibrium points: one to the right of E_{m}, which is a saddle point, and one to its far right, which is typically a unstable spiral. It is known from the theory of dynamical systems that the resting membrane potential is always stable and the triple-equilibria configuration creates a saddle-node bifurcation point for the threshold of firing, giving rise to an all-or-nothing firing mechanism for action potentials. If the initial voltage is to the right of the middle saddle-node, an action potential ensues, with a magnitude stretching passing the spiral focus at the least. In contrast, Fig. 6c shows that the HH model has only one equilibrium point which is the resting membrane potential E_{m}. The

corresponding firing mechanism is by the way of a Hopf bifurcation point. In theory, the action potentials are graded, not the all-or-nothing type as is supposed to be. That is, depending on how close the initial voltage is to the Hopf-point [the local minimum point of the reflection of the joint sodium-leakage *IV*-curve $I = -(f_{\mathrm{Na}}(V) + f_{\mathrm{L}}(V))$], the magnitude of an action potential can vary from nothing to full, a phenomenon of the so-called canard explosion (Benoit et al. 1981; Deng 2004).

There are three more noticeable differences about the firing mechanism. For our model, the stability of the resting potential is due to the voltage-gating *IV*-curve, whereas that for the HH model is critically dependent on the presence of the leakage current I_{L}. Second, the repolarization of the membrane for our model is due to the voltage-gating characteristics as the sodium channel is closing down when charged molecules return to the sodium gate pores. In contrast, the repolarization of the membrane by the HH model is carried out by the leakage channel. Third, our model predicts a persisting leaking current when the membrane is at the resting potential E_{m} as shown in the inset of Fig. 6a of which the primary components are the gating and the sodium currents, canceling out each other at the resting equilibrium. That is, the leakage is a consequence of the voltage-gating rather than a cause by the HH model shown in the inset of Fig. 6c.

As for the action potential at large, our model shows it tracks closely around the sodium-gating curve $I = -(f_{\mathrm{Na}}(V) + f_{\mathrm{G}}(V))$. Also the reversal of the potassium current (trajectory entering the left side of the potassium *IV*-curve) happens rather promptly after the potassium current has peaked. In contrast, the action potential of the HH model comes close only to two points of the *IV*-curves, the Nernst potentials E_{K}, E_{Na}, and with the rest playing little role to guide the trajectory. And its potassium current reversal is rather delayed after the potassium current has peaked and the membrane has started depolarization.

3.3 Voltage-Gating

One major difference lies in the inclusion of voltage-gating into our model. As shown in Fig. 6b, voltage-gating is active during the initialization phase and the termination phase of an action potential. Both are for the purpose of preventing (accidental) firing of action potentials, and yet once the firing threshold is exceeded, the gating conductance recedes, permitting the membrane to depolarize. In contrast, little can be deduced from the leakage conductance time course (which is a constant of time) from the HH model as shown in Fig. 6d. Figure 7 also shows more clearly what happens with the injection of an excitatory intracellular square pulse $I_{\mathrm{ext}}(t)$. Voltage-gating suppresses an action potential if the excitation fails to clear the threshold, approximately the maximum of the gating current. Otherwise an action potential is generated if the excitation clears the threshold. The figure also shows that taking away gating ($\bar{g}_{\mathrm{G}} = 0$), the membrane resting potential E_{m} is gone for losing its stability and the action potential is replaced by a perpetual limit cycle. It also shows that if the gating range is too close to the gating resting potential E_{G} (very small b_{G}), it is effectively the same as if there is no gating ($\bar{g}_{\mathrm{G}} = 0$). Also if the range of gating is far away (large b_{G}), the threshold for action potential can become too high for intracellular excitation to

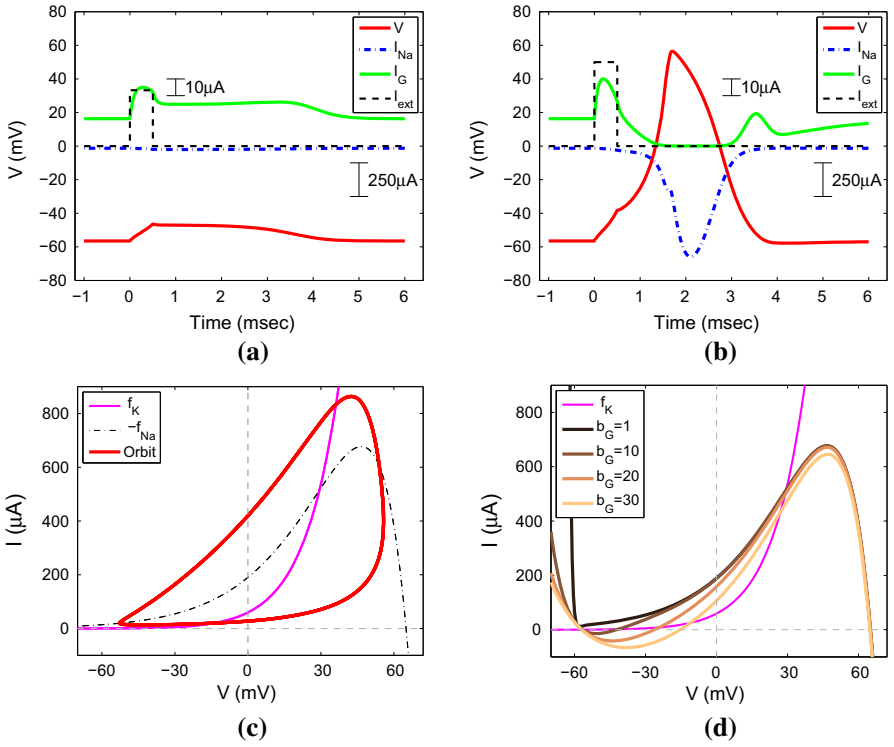


Fig. 7 (Color figure online) **a** With the same parameter values as from Fig. 4a for model (8) but forced by an excitatory square pulse I_{ext} just below the action potential threshold which is about the maximal value of the gating current. **b** The same plot except for an above-threshold square pulse. **c** A cyclic firing without the repolarizing resting potential if voltage-gating is absent ($\bar{g}_{\text{G}} = 0$). **d** The sodium-gating *IV*-curves with varying gating range b_{G}. Too small or too big a b_{G} is bad for action potential generation

clear. There is a 'Goldilocks Zone' from the resting gating potential for the range of gating.

3.4 Ion Current Activation

Figure 8 shows the effect of the sodium current activation range parameter b_{Na} on the generation of action potentials. If the sodium activation is too close to its Nernst resting potential (small b_{Na}), it requires a very long excitation time for action potentials to arise. If the activation is far away (large b_{Na}), it renders the voltage-gating useless. It also requires a 'Goldilocks Zone' from the resting sodium potential for the range of sodium activation.

3.5 Model Overfit

Another major difference lies in the arbitrariness of the functional forms for the voltage-dependent conductances of the HH model. For example, for the function α_n

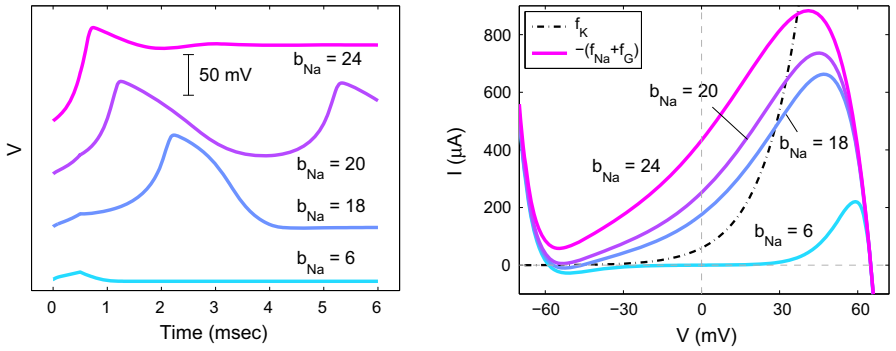


Fig. 8 (Color figure online) **a** With the same initial conditions, the same parameter values, and the same excitation forcing as in Fig. 7b but with varying sodium activation range parameter b_{Na}. **b** The corresponding *IV*-curve configurations. Too small or too big a b_{Na} is not desirable for action potentials

of the HH model, four parameters were actually required and fitted to have its final form obtained in Hodgkin and Huxley (1952a) because it is of this general form

$$\alpha_n = \frac{a_1 - V}{a_2(\exp(a_3 - a_4 V) - 1)}.$$

Similarly, the function β_n was fitted with two free parameters. Altogether, the HH model had 13 more parameters for its fitting than our model (of which only the b-parameters and the τ-parameters are not shared with the HH model). Because the functional forms of the HH model and their parameters are rather arbitrary, its fit to the data can be construed as 'overfit' as with the case where arbitrary polynomials can fit but not explain any data.

3.6 Traveling Action Potential

The term action potential in its original definition is referred to the uniform profile of the membrane voltage when an electrical pulse propagates from one end of the axon to another. The partial differential equations for such propagating pulse is derived by adding the axial diffusive current DV_{xx} to the total current at each location to the patch model (8). The derivation follows the same treatment as in Hodgkin and Huxley (1952a). The spatially continuous model is

$$\begin{cases} CV_t = DV_{xx} - [\bar{g}_{\mathrm{K}} n(V - E_{\mathrm{K}}) + \bar{g}_{\mathrm{Na}} m(V - E_{\mathrm{Na}}) + \bar{g}_{\mathrm{G}} h(V - E_{\mathrm{G}}) - I_{\mathrm{ext}}] \\ n_t = \tau_{\mathrm{K}}(\mathrm{e}^{(V-E_{\mathrm{K}})/b_{\mathrm{K}}} - n) \\ m_t = \tau_{\mathrm{NaG}}(\mathrm{e}^{(V-E_{\mathrm{Na}})/b_{\mathrm{Na}}} - m) \\ h_t = \tau_{\mathrm{NaG}}(\mathrm{e}^{-(V-E_{\mathrm{G}})/b_{\mathrm{G}}} - h) \end{cases} \tag{10}$$

with D being the axial diffusion coefficient of the axon. One can also derive a discrete version of the continuum model as follows

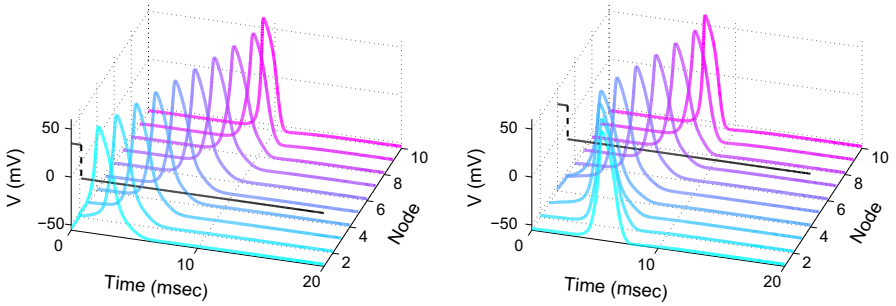


Fig. 9 (Color figure online) With the same parameter values as from Fig. 4a for model (11), and $d = 0.6$ m mho/cm^2, **a** shows the traveling pulse if the first node is injected with an intracellular current $I_{\text{ext}} = 35\ \mu$A for the first 0.8 msec. Plot **b** shows the same except that the excitatory current is injected at the fourth node

$$\begin{cases} CV_i' = -[\bar{g}_{\text{K}} n_i (V_i - E_{\text{K}}) + \bar{g}_{\text{Na}} m_i (V_i - E_{\text{Na}}) + \bar{g}_{\text{G}} h_i (V_i - E_{\text{G}}) \\ \qquad\qquad -I_{\text{ext}} + d_i (V_i - V_{i-1}) - d_{i+1} (V_{i+1} - V_i)] \\ n_i' = \tau_{\text{K}} (\text{e}^{(V_i - E_{\text{K}})/b_{\text{K}}} - n_i) \\ m_i' = \tau_{\text{NaG}} (\text{e}^{(V_i - E_{\text{Na}})/b_{\text{Na}}} - m_i) \\ h_i' = \tau_{\text{NaG}} (\text{e}^{-(V_i - E_{\text{G}})/b_{\text{G}}} - h_i) \end{cases} \tag{11}$$

with $i = 1, \ldots, n$, $d_1 = d_{n+1} = 0$, and $d_i = d$, a constant for $i = 2, \ldots, n$. The latter can also be used as a model for myelinated axon with i denoting the nodes of Ranvier ordered from one end of the axon to another. Figure 9 shows the phenomenon of saltatory propagation by the discrete model.

3.7 Spike Burst

We end this section with the phenomenon of spike burst (Arvanitaki and Chalazonitis 1949; Li and Jasper 1953; Frazier et al. 1967; Carpenter and Gunn 1970) for which the membrane potential of an excitable cell sustains a number of rapid spikes before hyperpolarizing and then polarizing near the resting potential of the membrane. Early models for bursting spikes include (Connor and Stevens 1971; Partridge and Stevens 1976; Morris and Lecar 1981; Chay and Keizer 1983; Hindmarsh and Rose 1984). We will not fit our model to any specific data per se but rather highlight the simple mechanisms and configurations for spike bursting. The basic template for spike burst requires the all-or-nothing firing mechanism shown in Fig. 6a. The key difference between the one-spike action potential and the many-spikes burst is the time adaptation constant for the potassium conductance τ_{K}. For the former, it has a slower timescale as shown in the figure where the sodium-gating reflection curve $I = -(f_{\text{Na}}(V) + f_{\text{G}}(V))$ strongly pulls the orbit to the left side of the potassium IV-curve $I = f_{\text{K}}(V)$. If we speed up the potassium adaptation by increasing its time constant, the orbit will come down faster toward the potassium IV-curve $I = f_{\text{K}}(V)$, and as a result oscillate around the spiral focus equilibrium point at the far right. That is, the middle saddle-node equilibrium can act to direct the orbit either to its left if the time constant τ_{K} is

small or to its right to form spike burst if $\tau_{\rm K}$ is large. Together, this forms the basic ingredients for spike burst.

But in order to sustain multiple bursts, another ion current is needed to drive the bursting-spiking orbit between the two regimes, being separated and redirected by the middle saddle-node point. The dynamics is into the spiking regime when an orbit is directed to the right of the point and into the quiescent regime when the orbit is directed to the left of the point. For illustration purpose, we will include the chloride ion Cl^- current to the membrane model. We will assume it has a negative Nernst potential $E_{\rm Cl} < 0$ above the resting membrane potential $E_{\rm m}$, an exponential activation conductance just like the other two ion species with moderate intrinsic conductance $\bar{g}_{\rm Cl}$ and a comparable activation voltage parameter $b_{\rm Cl}$. Last, we will assume that it is current-adapted and slowly so $0 < \tau_{\rm Cl} \ll 1$. The resultant system is as follows

$$\begin{cases} CV' = -[\bar{g}_{\rm K} n(V - E_{\rm K}) + \bar{g}_{\rm Na} m(V - E_{\rm Na}) + \bar{g}_{\rm G} h(V - E_{\rm G}) + I_{\rm Cl} - I_{\rm ext}] \\ n' = \tau_{\rm K}(e^{(V-E_{\rm K})/b_{\rm K}} - n) \\ m' = \tau_{\rm NaG}(e^{(V-E_{\rm Na})/b_{\rm Na}} - m) \\ h' = \tau_{\rm NaG}(e^{-(V-E_{\rm G})/b_{\rm G}} - h) \\ I_{\rm Cl}' = \tau_{\rm Cl}(\bar{g}_{\rm Cl} e^{(V-E_{\rm Cl})/b_{\rm Cl}}(V - E_{\rm Cl}) - I_{\rm Cl}) \end{cases} \tag{12}$$

We also note that for many types of neuron and excitable membranes their spike-burst dynamics are determined by sodium, potassium, and calcium ions (Ca^{++}) with the role of the sodium in the model above being replaced by the calcium (whose Nernst potential is usually higher still than that of the sodium ion) and the role of the chloride replaced by the sodium so that the region of spikes is always above the zero membrane potential $V = 0$. For mathematical mechanisms which generate spike bursts, the prototypical model above is nonetheless rather typical and illuminating. More specifically, Fig. 10a shows a simulation of such a spike burst. One can see that gating again plays an important role for this behavior, shutting down the burst when it is activated and permitting spiking when it recedes. Also, it shows that the addition of the slow chloride current increases during the spiking phase until sending the dynamics into the quiescent phase and decreases during the silent phase until sending the dynamics into the active phase.

Spike burst is a 3-dimensional phenomenon just like action potential is basically a 2-dimensional one. To demonstrate this point, let assume the sodium and gating dynamics is the fastest with $\tau_{\rm NaG} \sim \infty$ so that their conductances equilibriumize instantaneously with $m = e^{(V-E_{\rm Na})/b_{\rm Na}}$ and $h = e^{(V-E_{\rm G})/b_{\rm G}}$. This eliminates two differential equations from (12) above to obtain the 3-dimensional system below

$$\begin{cases} CV' = -[\bar{g}_{\rm K} n(V - E_{\rm K}) + \bar{g}_{\rm Na} e^{(V-E_{\rm Na})/b_{\rm Na}}(V - E_{\rm Na}) \\ \qquad\qquad + \bar{g}_{\rm G} e^{-(V-E_{\rm G})/b_{\rm G}}(V - E_{\rm G}) + I_{\rm Cl} - I_{\rm ext}] \\ n' = \tau_{\rm K}(e^{(V-E_{\rm K})/b_{\rm K}} - n) \\ I_{\rm Cl}' = \tau_{\rm Cl}(\bar{g}_{\rm Cl} e^{(V-E_{\rm Cl})/b_{\rm Cl}}(V - E_{\rm Cl}) - I_{\rm Cl}) \end{cases} \tag{13}$$

Figure 10b is the phase space of this system at the cross-section $I_{\rm Cl} = 0$. The V-nullcline and the n-nullcline clearly show the all-or-noting firing configuration, having

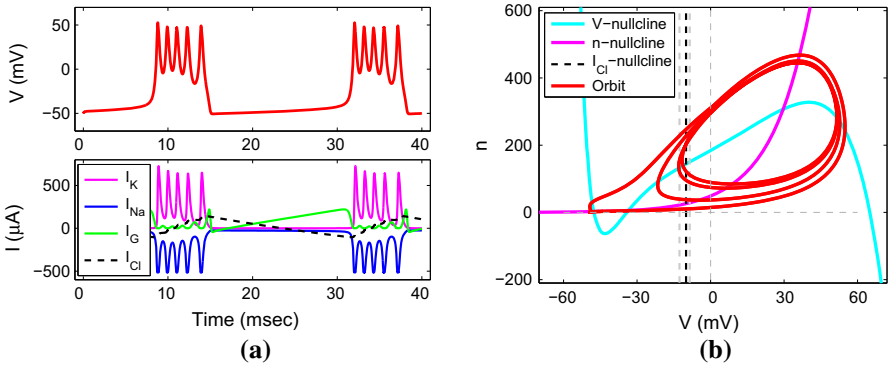


Fig. 10 (Color figure online) **a** Spike bursts of model (8) with the parameter values: $E_{\mathrm{K}} = -62$, $\bar{g}_{\mathrm{K}} = 0.015$, $b_{\mathrm{K}} = 16$, $E_{\mathrm{Na}} = 65$, $\bar{g}_{\mathrm{Na}} = 70$, $b_{\mathrm{Na}} = 20$, $E_{\mathrm{G}} = -50$, $\bar{g}_{\mathrm{G}} = 15$, $b_{\mathrm{G}} = 10$, $C = 1$, $\tau_{\mathrm{K}} = 3$, $\tau_{\mathrm{NaG}} = 80$, $E_{\mathrm{Cl}} = -10$, $\bar{g}_{\mathrm{Cl}} = 10$, $b_{\mathrm{Cl}} = 50$, and $\tau_{\mathrm{Cl}} = 0.02$. The gating current and the chloride current are magnified four times for a better view. **b** Exactly the same parameter values for the 3-D model (13). The V-nullcline shown is its intersection with $I_{\mathrm{Cl}} = 0$. *Dark dash line* is the projected I_{Cl}-nullcline with $I_{\mathrm{Cl}} = 0$ and the *light dash lines* mark the nullcline's effective range for the spike burst

three intercept points. It also shows the spikes lie to the right side of the middle saddle-node of the Vn-subsystem. As most part of the spikes lies to the right side of I_{Cl}-nullcline in which $I'_{\mathrm{Cl}} > 0$, I_{Cl} increases to become more positive. The outward current inhibits spiking, lowering the V-nullcline until the orbit is caught to the left side of the middle saddle-node point and the spikes are switched off. But the quiescent phase lies to the left side of the I_{Cl}-nullcline in which $I'_{\mathrm{Cl}} < 0$, I_{Cl} decreases to become more negative, changing it into an excitatory current to drive the membrane into its spiking regime again. This is the basic 3-dimensional blueprint for continuous bursts of spikes in all higher dimensions. For exactly the same parameter values, the lower-dimensional system (13) and the higher-dimensional counterpart (12) are all comparable except for possibly a different number of spikes. That is, if we project the higher-dimensional bursts of spikes from Fig. 10a to its Vn-space, it should look like the phase portrait Fig. 10b of the lower-dimensional counterpart.

4 Concluding Remark

Our mathematical modeling of the squid axon began with a model for the sodium-potassium ion exchanger pump which led to a justification of the Nernst potentials for both ion species and to the modeling of the ion currents by battery-driven conductors. Following upon the voltage-dependent conductance finding of Hodgkin and Huxley, we introduced the exponential activation model for both opening and closing of ion channels. This activation model unifies two seemingly different types of activation in potassium and sodium channels. We also introduced the exponential voltage-gating model for the sodium channel which together with the sodium activation model automatically led to N-shaped sodium-gating characteristics, solving the negative conductivity problem which has puzzled researchers of many generations. By incorporating Hodgkin and Huxley's time adaptation for channel conductances, our

model is capable of generating the all-or-nothing action potentials as a consequence to voltage-gating functional form. These voltage-gating-related properties seem to be novel to our best knowledge.

More interestingly, our model enables the following description on action potential generation, although most of it known from the HH model but now cast in the mechanistic assumptions for our model. Specifically, in the absence of external excitation the membrane is kept at rest by voltage-gating. When depolarizing excitation exceeds a threshold above the maximal voltage-gated current, the voltage-gating current drops exponentially to zero, allowing the membrane to depolarize which in turn opens up the sodium gates. The increased inward sodium current further depolarizes the membrane ($\mathrm{d}V/\mathrm{d}t > 0$) which in turn opens up not only more sodium gates but also potassium gates as well in the manner that their channel conductances grow exponentially with the depolarizing voltage. However, the inward sodium current must slow down as it is buffered by the potential difference between the membrane potential and its positive Nernst potential ($I_{\mathrm{Na}} = g_{\mathrm{Na}}(V)(V - E_{\mathrm{Na}})$), but in contrast the outward potassium current runs away exponentially from its negative Nernst potential. The outward potassium current must eventually catch up with the inward sodium current in magnitude so that the net ion and gating current becomes outward and the direction of the across-membrane voltage is reversed ($\mathrm{d}V/\mathrm{d}t \leq 0$). This reversal activates the closing of both ion gates, closing up them exponentially fast. This in turn speeds up the downfall of the membrane potential. The repolarizing potential may not pass its resting potential if the outward potassium current closes itself too quickly, giving rising to continuous spiking of some sort. Otherwise, the across-membrane potential must overshoot the voltage-gating potential ($V < E_{\mathrm{G}}$), in which case the already-activated voltage-gating conductance is near its intrinsic conductance. From this point on the system must converge to the membrane resting potential whether or not the voltage overshoots it. This is because of two interplays of the currents. One, the voltage-gating current becomes negative below its reversal potential. Two, the sodium and potassium currents become so much smaller than the small inward voltage-gating current. So the latter becomes dominating, effectively halting membrane hyperpolarization and then reversing the direction of the hyperpolarizing membrane potential ($\mathrm{d}V/\mathrm{d}t \geq 0$), and bringing the membrane to its polarized resting state yet again.

It seems that our approach to modeling excitable membranes works no worse than the HH model. It can both fit and explain, within our model hypotheses some important features of excitable membranes. Its modeling methodology can be easily adopted for other neuronal systems. Its dynamics are readily accessible by standard singular perturbation techniques and dynamical systems analyses. Its simplicity and accuracy should prove useful when the model is used as building blocks for large network simulations and studies. It should be considered as an alternative to the HH model with the acknowledgement that it definitely will not be the last.

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